Persistent Black–White Preterm Birth Disparities in U.S. Natality Public-Use Data, 2016–2024

N ≈ 33,511,275 births · CDC/NCHS Natality public-use files

Abstract / Thesis

Despite secular changes in overall U.S. birth outcomes over 2016–2024, the non-Hispanic Black–White preterm disparity remained large and did not close; racial disparity in preterm birth is not explained by the available socioeconomic and care-timing covariates in the public-use file. From 2016 to 2024, the non-Hispanic Black–White preterm birth rate ratio widened (RR 1.52 → 1.56), with Black rates 14.86% vs White 9.51% in 2024. In multivariable logistic models, non-Hispanic Black identity retained elevated odds of preterm birth (OR 1.49, 95% CI 1.46–1.53) after adjustment for education, prenatal care timing, tobacco, obesity, diabetes, hypertension, plurality, payment source, age, and year. The absolute prenatal-care initiation gradient in preterm rates (15.09 percentage points between no care and first-trimester care) exceeded the education gradient (2.96 pp between lowest and highest education).

Key quantities

- Preterm rate: 9.85% → 10.41%
- Black–White preterm RR: 1.52 → 1.56
- Latest Black vs White preterm rates: 14.86% vs 9.51%
- Adjusted OR (NH Black, preterm): 1.49 (95% CI 1.46–1.53)
- LBW: 8.17% → 8.52%; Cesarean (latest) 32.38%; NICU (latest) 9.88%
- Education / PNC absolute preterm gradients: 2.96 / 15.09 pp

Data & methods

U.S. Natality public-use fixed-width microdata; obstetric-estimate preterm; complete-case rates; multivariable logistic regression (statsmodels). Geography unavailable post-2004. Full markdown paper and numerical tables are in the repository.

Limitations

No state/county identifiers; observational associations only; residual confounding; item nonresponse; low-risk cesarean is a limited proxy.

Reproducibility

https://github.com/Kesmitij/cdc-natality-birth-outcomes
scripts/download_natality.py → process_all_years.py → run_analysis.py → build_paper.py

Paper body (excerpt from Markdown source)

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Abstract

Thesis. Despite secular changes in overall U.S. birth outcomes over 2016–2024, the non-Hispanic Black–White preterm disparity remained large and did not close; racial disparity in preterm birth is not explained by the available socioeconomic and care-timing covariates in the public-use file. From 2016 to 2024, the non-Hispanic Black–White preterm birth rate ratio widened (RR 1.52 → 1.56), with Black rates 14.86% vs White 9.51% in 2024. In multivariable logistic models, non-Hispanic Black identity retained elevated odds of preterm birth (OR 1.49, 95% CI 1.46–1.53) after adjustment for education, prenatal care timing, tobacco, obesity, diabetes, hypertension, plurality, payment source, age, and year. The absolute prenatal-care initiation gradient in preterm rates (15.09 percentage points between no care and first-trimester care) exceeded the education gradient (2.96 pp between lowest and highest education).

Data. We analyze U.S. (not territory) Natality public-use microdata from the National Center for Health Statistics / CDC for 2016–2024 (N ≈ 33,511,275 births after U.S. resident filters), using official fixed-width layouts from NCHS User Guides.

Methods. We construct preterm birth (obstetric estimate <37 weeks), low and very low birth weight, cesarean delivery, NICU admission, low 5-minute Apgar, neonatal antibiotics, and infection indicators, with explicit handling of “not stated” codes. We report year trends, stratified rates (race/ethnicity, education, prenatal care timing, payment, clinical factors), Black–White rate ratios, and multivariable logistic models.

Results. Overall preterm rates moved from 9.85% to 10.41% across the window. The non-Hispanic Black–White preterm rate ratio was 1.52 at the start and 1.56 at the end (absolute gap 5.35 percentage points in the latest year: Black 14.86% vs White 9.51%). Adjusted OR for NH Black (vs NH White) preterm birth was 1.49 (95% CI 1.46–1.53).

Conclusions. Measured socioeconomic and care-timing covariates in the public-use file do not eliminate the Black–White preterm association. Geography is unavailable post-2004, limiting place-based inference.

Keywords. preterm birth; birth certificates; racial disparities; CDC Natality; low birth weight

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1. Introduction

Infant and maternal birth outcomes remain central indicators of population health in the United States. Preterm birth and low birth weight concentrate later childhood morbidity and drive neonatal intensive care utilization. Racial and socioeconomic disparities in these outcomes have been documented for decades in vital statistics and clinical cohorts.

This paper uses the **complete national public-use Natality microdata** (birth certificates) to (i) describe recent trends, (ii) quantify disparities along race/ethnicity, education, and prenatal care gradients, and (iii) estimate multivariable associations for primary outcomes. Critically, the **thesis is not pre-specified as a narrative of improvement or worsening**; it is derived from the estimated tables after analysis (see Abstract).

2. Data and Methods

2.1 Data source

Official CDC/NCHS Natality **U.S.** public-use files (`NatYYYYus.zip`) from the [Vital Statistics Online Data Portal](https://www.cdc.gov/nchs/data_access/VitalStatsOnline.htm), years 2016–2024. Territory files are excluded. Field positions follow NCHS User Guides (see `docs/DATA_PROVENANCE.md` and `src/natality/layouts.py`). Download URLs and SHA-256 checksums are recorded in `data/raw/download_manifest.json`.

2.2 Sample

Births to U.S. residents (`RESTATUS` 1–3). Foreign residents are excluded. Annual counts are checked against published NCHS U.S. totals (2% relative tolerance) in process metadata.

2.3 Outcomes and covariates

Documented in `docs/VARIABLES.md`. Primary outcomes: preterm (OE <37 weeks), LBW (<2500 g), VLBW (<1500 g), cesarean, NICU, low Apgar5, infection composite, neonatal antibiotics. Stratifiers: maternal age, education, race/ethnicity (single-race / Hispanic categories via `MRACEHISP`), prenatal care initiation, tobacco, obesity, diabetes, hypertension, plurality, payment source, year.

Missing and “not stated” codes are set to null and excluded from complete-case denominators; they are never coded as event-absent.

2.4 Statistical analysis

1. **Descriptive trends:** annual complete-case rates per 100 births. 2. **Disparities:** rates by race/ethnicity; Black–White rate ratios and absolute differences by year; education and prenatal care gradients. 3. **Multivariable logistic regression** for primary outcomes with maternal age, race/ethnicity (ref: NH White), education (ref: HS/GED), prenatal care timing (ref: first trimester), tobacco, obesity, diabetes, hypertension, plurality, payment (ref: private), and centered year. Complete-case on model variables. When the stacked microdata exceed ~1.5M rows, models use documented outcome-stratified samples for computational feasibility; descriptives use the full extract.

Software: Python 3.12, pandas/polars, statsmodels, matplotlib. All scripts are in the public repository.

3. Results

3.1 Cohort

Approximately **33,511,275** U.S. resident births across 2016–2024.

3.2 National trends

year	n_births	preterm_rate	lbw_rate	vlbw_rate	cesarean_rate	nicu_rate	low_apgar5_rate
2016	3945875	9.847197952730061	8.165690141745596	1.3982493846662456	31.916267855222824	8.739217866209444	2.0192587
2017	3855500	9.933862670758298	8.27704517951729	1.4051921636299356	31.98098589045529	8.980345102942108	2.012524317
2018	3791712	10.022603685997458	8.281082564836174	1.3769867873035648	31.876581724081888	9.154775285496097	1.996442
2019	3747540	10.228232154043086	8.313451955622414	1.381804719819481	31.677132731633613	9.3289681166109	2.067896215
2020	3613647	10.093324350092725	8.242348373016643	1.3384219549280327	31.805661541324113	9.339335429449978	2.102355
2021	3664292	10.48750443831426	8.520495028491153	1.3808123058399147	32.079113978501674	9.615074490690098	2.1829465
2022	3667758	10.383055027070299	8.603218539345464	1.3608228406272922	32.143429151583796	9.494780848067116	2.190895
2023	3596017	10.405315337252583	8.579300003172746	1.3590485324379882	32.334204190131004	9.800965765821987	2.244689
2024	3628934	10.407129083727114	8.52098806902815	1.3274623056470032	32.37914296988612	9.881636596105833	2.26679365

Preterm moved from **9.85%** to **10.41%**; LBW from **8.17%** to **8.52%**. Latest cesarean and NICU rates were **32.38%** and **9.88%**, respectively.

3.3 Black–White preterm disparity

year	white_rate	black_rate	rate_ratio	rate_diff_pp
2016	9.065912675869507	13.761502127160481	1.517938967555809	4.695589451290974
2017	9.080284792601551	13.917757658193771	1.5327446193685141	4.83747286559222

2018	9.113171040151755	14.124197294712392	1.5498663673141366	5.011026254560637
2019	9.283260417876548	14.389270294625376	1.5500233373736139	5.1060098767488284
2020	9.122401971517752	14.349626430705843	1.573009660778893	5.227224459188092
2021	9.52160730643351	14.743568887213181	1.548432781643003	5.221961580779672
2022	9.45876777302094	14.57871665571266	1.5412913188645196	5.11994888269172
2023	9.46838153162179	14.648030516057846	1.547046923187184	5.1796489844360565
2024	9.505900956195532	14.857628701685169	1.5629900595589115	5.351727745489637

The rate ratio was **1.52** in the first year and **1.56** in the last year; the absolute difference in the latest year was **5.35** percentage points (Black **14.86%**, White **9.51%**).

3.4 Education and prenatal care gradients

Education (preterm):

level	n	events	rate_per_100
Less than HS	3996122	458223	11.466691957853138
HS / GED	8645337	962277	11.13058981969124
Some college	6276233	678645	10.812935083831336
Associate	2757497	286839	10.402150936156957
Bachelor's	7042824	599281	8.509100894754718
Graduate	4268722	364657	8.542533339018094
Missing	498174	63980	12.842902279123358

Prenatal care timing (preterm):

level	n	events	rate_per_100
1st trimester	25263841	2492341	9.8652497060918
2nd trimester	5363779	517516	9.648346809217903
3rd trimester	1480045	115500	7.803816775841275
No PNC	633524	158092	24.95438215442509
Missing	743720	130453	17.54060667993331

Absolute education and prenatal-care spans were approximately **2.96** and **15.09** percentage points, respectively.

3.5 Multivariable models

Adjusted odds ratio for non-Hispanic Black (vs non-Hispanic White) preterm birth: **1.49** (95% CI **1.46–1.53**). Full coefficient tables are in `results/models/`.

term	or	ci_low	ci_high	pvalue
const	0.18319384285589163	0.17571989564304738	0.1909856817151881	0.0
maternal_age	1.0169131208020064	1.0155631453276652	1.0182648907819771	3.475265905380662e-135
year_c	1.0032060730561678	1.0004391710654854	1.0059806274327696	0.023113474394255863
tobacco	1.4742604574093432	1.430284535865435	1.5195884747266044	2.8569661226882513e-139
obesity	1.0153002709725705	0.9993622807870648	1.031492442785737	0.05997975657159842
diabetes	1.4727319336171718	1.4387032180003345	1.5075655084099975	4.3554978663672045e-231
hypertension	2.995692694351295	2.938719745669368	3.053770177375532	0.0
multiple	18.451895564690226	17.825659667082565	19.10013184863852	0.0
race_eth_Hispanic	1.0969261501657819	1.0762992829606013	1.117948323451191	1.277670065930206e-21
race_eth_NH AIAN	1.134487846972185	1.0478560497811915	1.228281952655942	0.0018496404087551491
race_eth_NH Asian	1.1240640494673841	1.0892612276888218	1.1599788509740052	3.1428964162714795e-13
race_eth_NH Black	1.4936593076895015	1.46276147474016	1.5252097939233678	9.6727448009108e-310
race_eth_NH Multiracial	1.1249881685730931	1.0732781817999961	1.1791895157198722	9.316528760390463e-07
race_eth_NH NHOPI	1.2375963738124511	1.0812325685799011	1.4165729270303076	0.00197956846995132
education_Associate	0.877			

4. Discussion

The organizing empir